

SaaS Financial Performance, Unit Economics and Revenue Leakage Analysis

Executive Summary

This project evaluates portfolio revenue durability, contract concentration risk, and churn-driven financial leakage for a SaaS subscription business managing \$5.47M in Annual Recurring Revenue (ARR) across 7,043 active accounts.

By combining an optimized Python/Pandas data pipeline with an in-memory ANSI SQL aggregation engine (DuckDB), this project bridges technical data engineering and strategic Financial Planning and Analysis (FP&A). The analysis pinpoints core operational bottlenecks—isolating \$1.67M in annual revenue leakage—and delivers C-suite executive recommendations to stabilize cash flows and improve unit economics.

Technical Stack and Architecture

- Language: Python 3.10+
- Data Manipulation and Vectorization: pandas, numpy (Named aggregations, pd.cut binning, SIMD-accelerated array math)
- SQL Aggregation Engine: duckdb (Zero-copy in-memory querying over DataFrame memory pointers)
- Environment: Visual Studio Code / Jupyter Notebook

Key Financial Findings

1. High Revenue Concentration and Contract Exposure

- Portfolio Scale: Total Portfolio ARR stands at \$5.47M derived from an MRR baseline of \$456.12K.

- Contract Risk: 56.41% (\$3.09M ARR) of total recurring revenue is tied to short-term Month-to-month contracts, exposing over half of company cash flows to immediate cancellation risk.

2. Severe Financial Leakage (\$1.67M ARR Loss)

- Revenue Churn: The company suffers an annual financial leakage of \$1.67M ARR (30.50% of total revenue) driven by 1,869 churned accounts.
- Premium Tier Decay: Churned subscribers generate a higher average monthly billing (\$74.44 Avg MRR) than retained subscribers (\$61.27 Avg MRR), proving that churn disproportionately impacts high-tier, valuable customer segments.

3. Lifecycle and Onboarding Bottlenecks

- First-Year Friction: The 0–1 Year (0–12m) cohort exhibits a critical 47.44% churn rate, indicating that nearly half of new subscribers cancel before the company recovers its Customer Acquisition Cost (CAC).
- LTV Expansion: Retention stabilizes past 24 months. The 4–6 Years cohort accounts for 36.30% (\$1.99M ARR) of portfolio revenue with an elite 9.51% churn rate.

4. Root Cause Isolation via In-Memory SQL Engine (DuckDB)

- Contract Leakage Concentration: 86.86% (\$1.45M) of all lost ARR originates directly from the Month-to-month segment (42.71% segment churn rate).
- Multi-Year Protection: Two-Year contracts restrict annual loss to just \$49.98K (2.83% churn rate), proving the stabilizing power of long-term contract lock-ins.

Executive Summary Table (SQL Output)

Contract	Total Customers	Churn Count	Churn Rate (%)	Contract ARR (\$)	Financial Leakage ARR (\$)	Avg MRR (\$)
Month-to-month	3,875	1,655	42.71 %	\$3,087,529.80	\$1,450,165.20	\$66.40
One year	1,473	166	11.27 %	\$1,149,799.20	\$169,421.40	\$65.05
Two year	1,695	48	2.83%	\$1,236,070.20	\$49,983.60	\$60.77

Strategic Recommendations for C-Suite Leadership

1. **Incentivize Annual Contract Migration:** Introduce a targeted 5–10% discount for converting Month-to-month subscribers into 1-Year or 2-Year commitments. Converting just 20% of monthly accounts would immediately safeguard ~\$290K+ in ARR.
2. **Overhaul Early Onboarding (0–12m):** Reallocate Customer Success resources toward first-year onboarding programs to reduce the 47.44% initial churn rate, accelerating CAC payback recovery.
3. **Address High-Tier Price Sensitivity:** Conduct a feature-utilization audit on premium packages (\$74.44+ MRR) to fix potential value-realization gaps causing top-tier account churn.